

MATH 195: 10/6

LOGARITHMS

We write $\log_a$ to refer to the *logarithm* function with positive *base* $a \neq 1$. It is common to write $\log_a x$ instead of $\log_a(x)$ for the output when the meaning is clear. Logarithms were invented to ease calculations. This is possible due to the following properties of logarithms.

Rules for logarithms.

- $\log_a a = 1$
- $\log_a(xy) = \log_a x + \log_a y$
- $\log_a(x/y) = \log_a x - \log_a y$
- $\log_a(x^y) = y \log_a x$

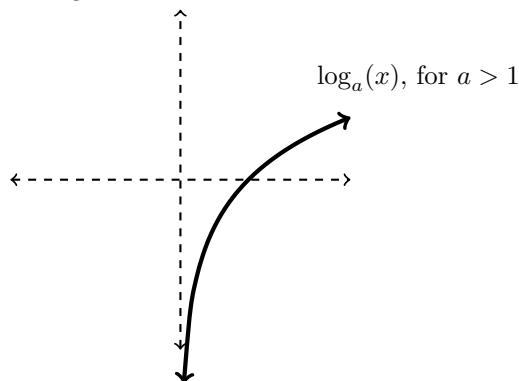
Briefly, you can remember these last three rules as: an operation inside the logarithm becomes a step less complicated outside the logarithm.

These rules may remind you of the rules for exponentials. Except, those go in the opposite direction; an operation inside the exponent becomes a step more complicated outside the exponential. The explanation for this is that logarithms and exponential functions are inverses of each other.

Inverse property of logarithms and exponentials.

- $\log_a(a^x) = a^{\log_a x} = x$
- $\log_a x = y \iff x = a^y$

This fact tells us how the graphs of logarithms look.



Vertical asymptote at $x = 0$. Increasing and concave down everywhere on its domain.

When graphing a geometric transformation of a logarithm, you need to know (1) where the vertical asymptote is and (2) what the orientation is.

Notation for special bases.

- The mathematically most important base for a logarithm is base e . We write $\ln$ for $\log_e$, and call it the *natural logarithm*.
- Depending on the source, people will write $\log$ (with no base specified) to mean different things. Pure mathematicians mean $\log_e$, computer scientists often mean $\log_2$, and many undergrad textbooks mean the *common log* $\log_{10}$. Because of this disagreement, I strongly discourage you from writing $\log$. Always specify the base.

PRACTICE PROBLEMS

The first batch of questions are about understanding logarithms' use as a calculational tool.

- (1) Explain why $\log_a(1) = 0$ for any base a .
- (2) Explain why $\log_{10}(10 \cdots 0)$ outputs the number of 0s in the input.
- (3) Explain why $\log_{10}(x)$ is between 0 and 1 if x is between 1 and 10.
- (4) Explain why $\log_{10}(3.54 \times 10^7) = 7 + \log_{10}(3.54)$.
- (5) More generally, explain why if you write a number in scientific notation as $D.d_1d_2 \cdots \times 10^k$ then

$$\log_{10}(D.d_1d_2 \cdots \times 10^k) = k + \log_{10}(D.d_1d_2 \cdots).$$

- (6) Explain how, knowing the values of $\log_{10}(x)$ for inputs x between 1 and 10 you can compute any common logarithm.
- (7) Describe an algorithm for multiplying two numbers using a table of outputs of the common logarithm for inputs between 1 and 10.
- (8) Describe an algorithm for finding square roots using a table of outputs of the common logarithm for inputs between 1 and 10.

The second batch of questions are about using the the inverse property of logarithms and exponential functions to solve algebraic problems.

- (9) Explain why $\log_a$ has its x -intercept at $x = 1$ for any base a .
- (10) Find the x -intercept of $a(x) = 2\log_3(x + 1) - 4$.
- (11) Find a formula for the inverse a^{-1} of a .
- (12) Find the x -intercept of $b(x) = e^{3-2x} - 2$.
- (13) Find a formula for the inverse b^{-1} of b .

The third batch of questions are about understanding the graphs of logarithms.

- (14) Use what you know about exponential functions and the domain and range of inverses to explain the domain and range of $\log_a$.
- (15) Must a geometric transformation of a logarithm always have an x -intercept? Explain. Same question for y -intercept.
- (16) Sketch a graph of $c(x) = \log_{4000}(x - 3)$. Explain why its vertical asymptote is at $x = 3$.
- (17) If you are asked to sketch a graph of $d(x) = \ln(mx + b)$ how would you determine where its vertical asymptote is? Give a general answer in terms of the parameters m and b .
- (18) Sketch a graph of $f(x) = -3\log_3(4 + x) + 1$. Determine the domain and range. Is it increasing or decreasing? Concave up or concave down?
- (19) Sketch a graph of $g(x) = 3\log_4(2 - 3x)$. Determine the domain and range. Is it increasing or decreasing? Concave up or concave down?
- (20) Sketch graphs of all four orientations for reflections of $\log_a(x)$ and write a formula for each.
- (21) When we looked at exponential functions, one fact we looked at is that for them a horizontal translation is equivalent to a vertical scaling. Explain why for logarithms a vertical translation is equivalent to a horizontal scaling.

[Hint: One approach is graphical, using what you know about exponentials and how graphs of an inverse relate to the original function. Another approach is algebraic, using rules for logarithms. Try to give an explanation using each approach.]